

PROBLEM STATEMENT: DESIGN A QUANTUM RESISTANT ENCRYPTION SYSTEM FOR SECURE COMMUNICATION IN MILITARY OPERATION

ABSTRACT:

This project presents a **hybrid secure communication system** that combines quantum key distribution with classical encryption to ensure highly secure data transmission. The system uses the BB84 protocol to generate a shared secret key between sender and receiver based on quantum principles. This key is then used with Fernet, a symmetric encryption method that internally uses the Advanced Encryption Standard, to encrypt the message.

The encrypted message is converted into a QR code for efficient and secure transmission. A receiver application scans the QR code, retrieves the encrypted data, and decrypts it using the same quantum-generated key. The system also includes emergency communication features for rapid predefined message transmission.

This approach ensures enhanced security by combining the **unconditional security of quantum key distribution** with the **practical reliability of classical encryption**, making it suitable for secure communication applications.

ALGORITHM:

1. Quantum Key Generation (BB84)

The BB84 protocol is used to generate a secure key based on quantum mechanical principles.

Steps:

- Generate random bits and random bases (+ or ×).
- Encode bits into qubits:
 - Normal basis → direct encoding
 - Diagonal basis → Hadamard transformation
- The receiver randomly selects bases and measures the qubits.
- Compare the bases:
 - If the bases match → retain the bit
 - If the bases do not match → discard the bit
- The final shared key is formed from the matching bits.
- Output: Secure binary key

2. Encryption using Fernet

- Convert the generated key into a 32-byte format.
- Initialize the Fernet cipher.
- Encrypt the message using the generated key.
- Produce the encrypted output.

- Output: Secure encrypted message

3. QR Code Transmission

- Convert the encrypted message into a QR code.
- Transmit or display the QR code to the receiver for secure transfer.

4. Receiver Process

- Scan the QR code using a camera interface.
- Extract the encrypted message from the QR data.
- Decrypt the message using the corresponding key.
- Display the original message.

UNIQUENESS OF THE PROJECT

1. Hybrid Security Model

Combines quantum key distribution (BB84) with classical encryption (Fernet/AES). Most existing systems rely on a single approach, whereas this system integrates both for enhanced security.

2. Quantum-Based Key Distribution

The key is not transmitted but generated using quantum principles, thereby eliminating the risk of interception during transmission.

3. Eavesdropping Detection Capability

In the BB84 protocol, any interception alters the quantum state, making unauthorized access detectable. This behavior is simulated within the system.

4. Practical Implementation Layer

While many quantum cryptography solutions remain theoretical, this system provides a practical implementation through:

- User interface using Streamlit
- QR-based communication
- Real-time interaction

5. Emergency Communication Feature

Supports predefined rapid-response messages such as STOP, ALERT, and RETREAT, enabling efficient communication in critical scenarios.

6. Secure and User-Friendly Design

Integrates advanced security mechanisms with a simple and intuitive QR-based interface for ease of use.

OUTPUT:

Sender side:

Deploy

Quantum Secure Sender

Emergency Signals

STOP

ALERT

RETREAT

AWARE

LOCATION

HELP

TARGET

SECURE

Selected Signal: all are safe

Or Enter Custom Message

all are safe

Generate Quantum Key

Quantum Key: 1110001011

Generate QR

Deploy

Generate QR

Encrypted Successfully



Scan this QR at Receiver

Encrypted Backup

gAAAAABpu40lqwClwGnKa2_TE5HwgT-YCePMawH-Wb8az1FC6wLBElyniPF5yy8Mrl5-_tFC9jQd-OafqzGRvNHktCQBzzfW==

RECEIVER SIDE:

Deploy



✕ Clear photo

QR Code Detected ✓

Encrypted Message

gAAAAABpu40lqwClwGnKa2__TE5HwgT-YCePMawH-Wb8az1FC6wLBElyniPF5yy8Mril5-__tFC9jQd-OafqzGRvNHktCQBzzfFw==

Deploy

✕ Clear photo

QR Code Detected ✓

Encrypted Message

gAAAAABpu4XgKpYUT5QjbelBR6G2MtfXSctnC3eKsLu9m_7ls_td0TgtEz7Po7v4wUvyOJTWca6h7HKj1Vd5Si2dOKD9sNIV3g==

Or Paste Encrypted Message

Enter Quantum Key

10100110010

Decrypt Message

all are safe